Extraocular Muscles

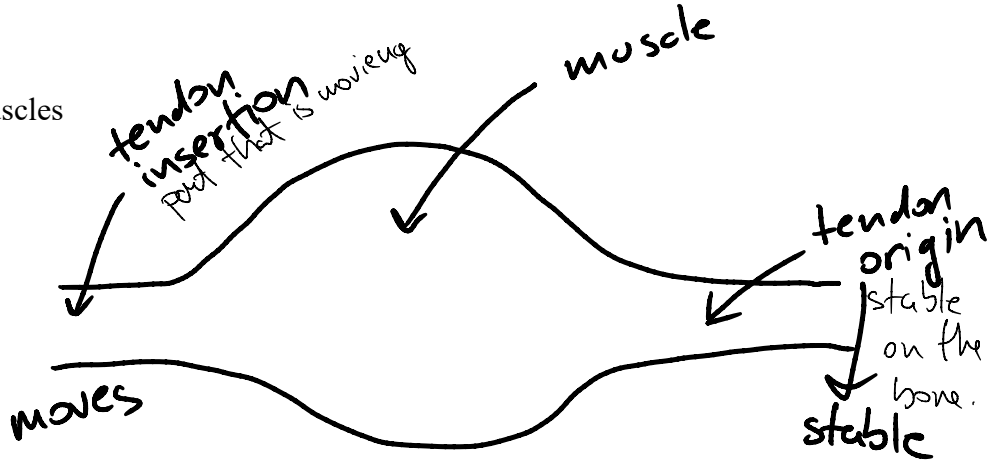
Images used: 11.10, 10.18, 11.17, 11.18, 11.15, 11.16, 11.19, 11.20, 13.14, 13.17

I. Macroscopic anatomy of the extraocular muscles

A. Six extraocular muscles:

1. Medial rectus (MR)
2. Lateral rectus (LR)
3. Superior rectus (SR)
4. Inferior rectus (IR)
5. Superior oblique (SO)
6. Inferior oblique (IO)

B. Origin of the rectus muscles



1. Common tendinous ring aka annulus of Zinn

2. The origin of the MR and SR also attach to the dural sheath of the optic nerve
   tissue around optic nerve
   a. Clinical correlate: Retrobulbar optic neuritis

C. Insertions of the rectus muscles

1. The four rectus muscles insert into the anterior globe

2. Tendons pierce Tenon capsule and merge with the scleral fibers.

D. Extraocular muscle anatomy

1. Medial rectus Largest muscle (muscle we use the most)

   a. Origin: common tendenist ~~(AI CHECK)~~ 2.2 ? CTR + ON SHEATH

   b. Insertion: antinior globe

   c. Innervation: inferior divison of CN 3 (AI CHECK) accolomotorius

2. Lateral rectus

   a. Origin: common "tendinost" ring = CTR + Greater wing (GW) of sphenoid

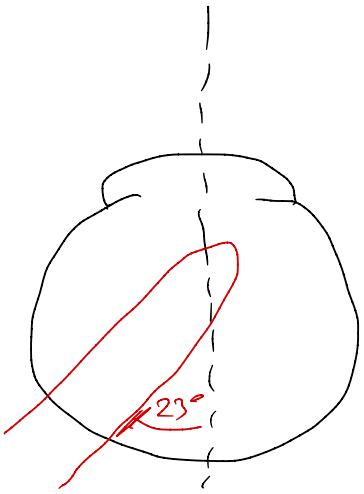
   b. Insertion: antinior globe

   c. Innervation: CN 6, abducens nerve

3. Superior rectus

   a. Origin: CTR + optic nerve sheath

   b. Insertion: anterior globe



   c. Innervation: superial division of CN 3 (oculomotorius)

4. Inferior rectus

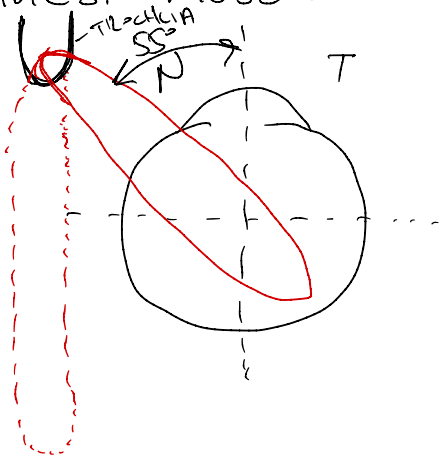
   a. Origin: CTR

   b. Insertion: anterior + globe @ 23° angle towards nasal side

   c. Innervation:

      inferior division of CN3

5. Superior oblique muscle longest and thinnest muscle



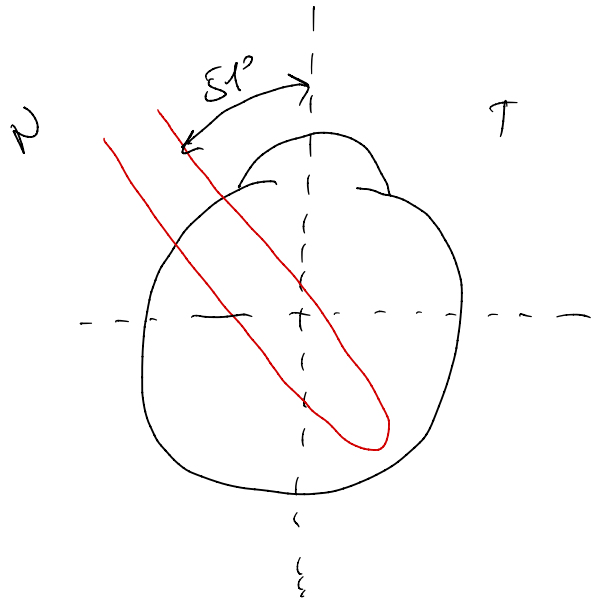
a. Pathway

b. Origin:
Trochlear

c. Insertion:
superior postinor lateral globe

d. Innervation:
CN4, trochlear (trohleoris)

6. Inferior oblique



   a. Pathway

   b. Origin:

maxillary bone (stay still)

   c. Insertion:

postirior lateral inferior globe

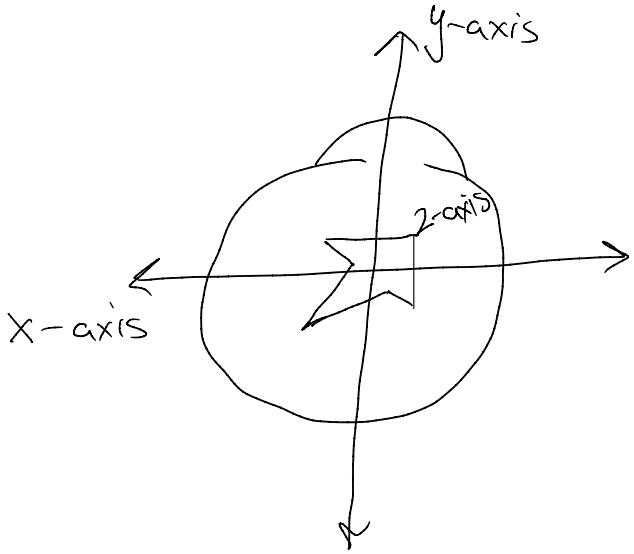
   d. Innervation:

inferior division of CN 3

WHY ARE YOU POKING ME IN THE EYE

II. Extraocular muscle movement definitions

A. Fick's axes DRAW



1. X axis
   a. Horizontal
   b. Runs nasal to temporal

2. Y axis
   a. Horizontal
   b. Runs anterior to posterior

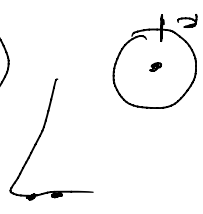
3. Z axis
   a. Vertical
   b. Superior to inferior

B. Ductions

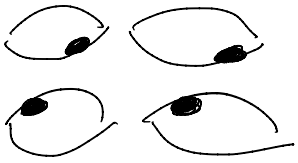
1. Rotation around the vertical z axis
   a. Medially: adduction
   b. Laterally: abduction

2. Rotation around the horizontal x axis
   a. Elevation: supraduction / elevation
   b. Depression: infraduction / depression

3. Rotation around the horizontal y axis relativly to the top of the eye
   a. Rotation nasally: intorsion / incyclorotation
   b. Rotation temporally: extorcion / excyclorotation



C. Position of gaze

1. Primary gaze: looking straight ahead, not accomodated 
2. Secondary gaze: vertical OR horizontal 
3. Tertiary position: vertical AND horizontal 

III. Extraocular muscle movements

A. Movement from primary position

1. Horizontal rectus muscles only have one action

a. Lateral rectus contraction abduction

b. Medial rectus contraction adduction

2. Vertical rectus muscles

a. Superior rectus

(1) Primary action: x-axes (pulling)
elevation

(2) Secondary actions: y-axes (pulling at 23°-diagonaly)
intorsion
adduction

b. Inferior rectus

(1) Primary action: x-axes
depression

(2) Secondary actions: y-axes
extorsion
2-axis adduction

3. Oblique muscles

   a. Superior oblique

      (1) Primary action: y-axis (pulling 55° and more, posf. lat.)
      INTORSION

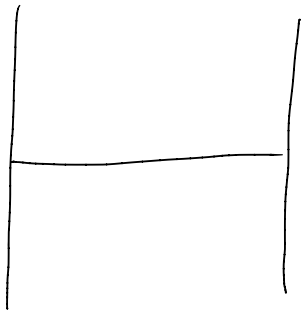
      (2) Secondary action: x-axis (pull pos. lat. sup. portion 55°)
      depession + abduction

   b. Inferior oblique

      (1) Primary action: extorsion (med. pos. lat. infir.)
      y-axes

      (2) Secondary action: x-axes
      elevation + 2-axes abduction

B. Movement from secondary positions



1. Vertical rectus muscles

   a. Abduction

      (1) There will be no torsional movement, and the eye is already abducted due to the LR.

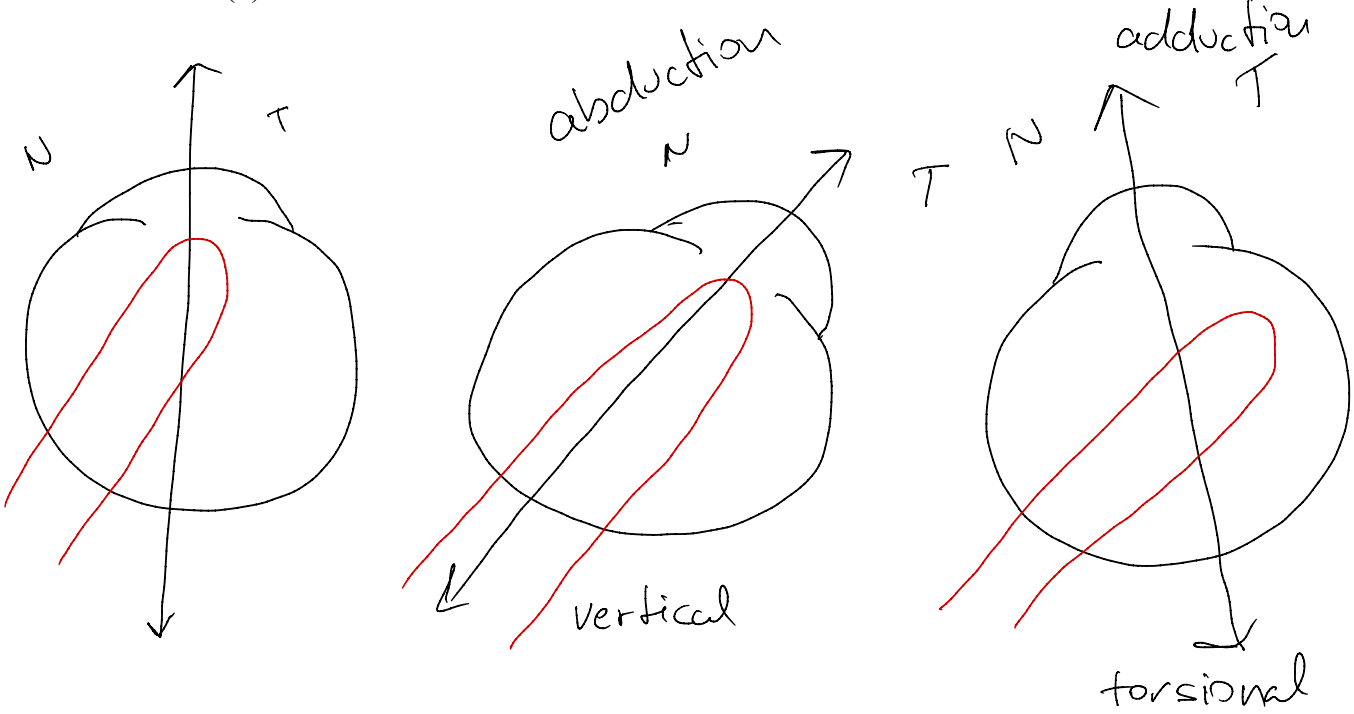
      (2) SR will cause

      (3) IR will cause

   b. Adduction

      (1) SR will cause

      (2) IR will cause



2. Oblique muscles Parallel axis ↑↓ "perpendicular" ⟳⟲

   a. Abduction

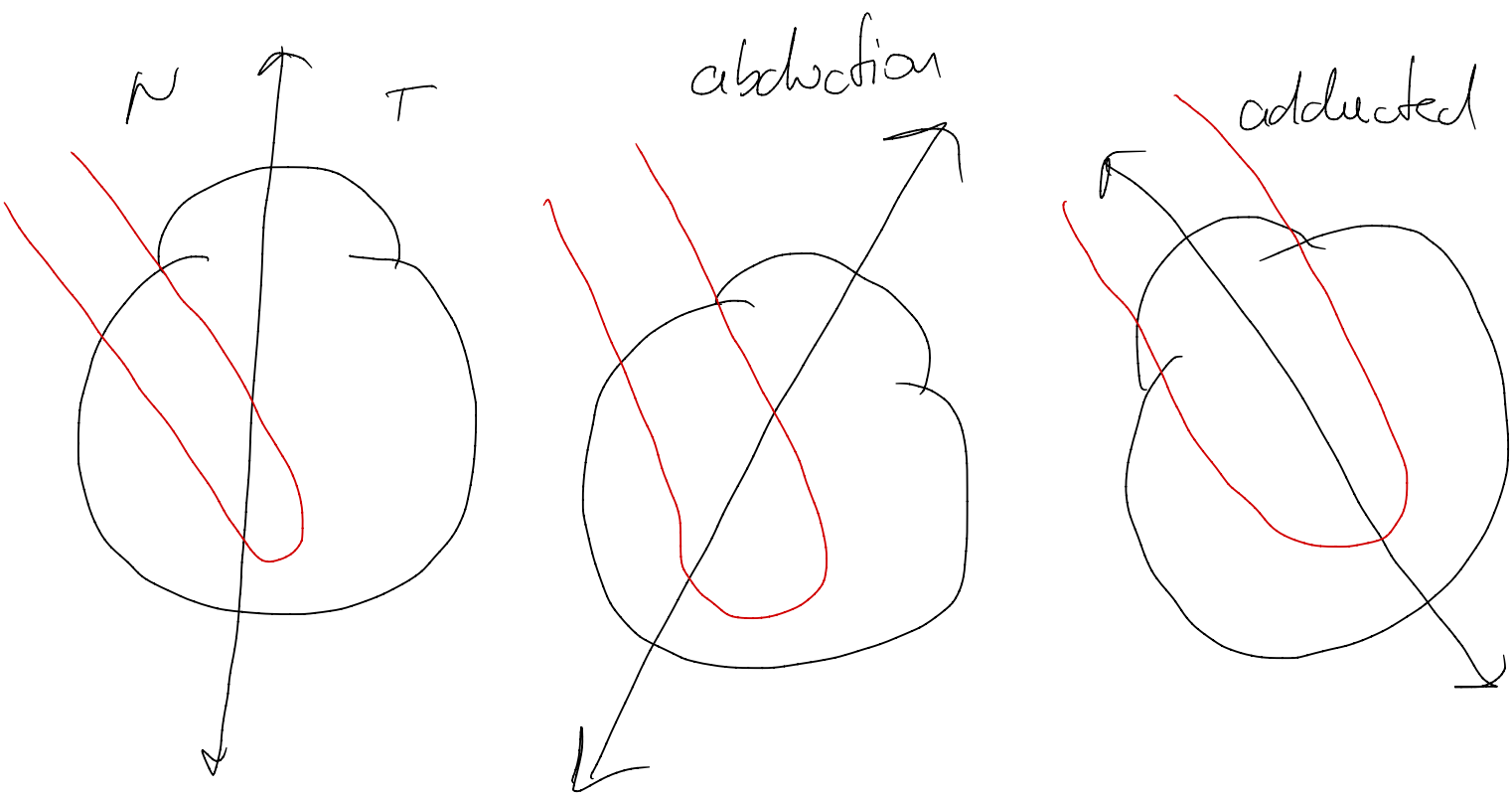
      (1) SO will cause intorsion

      (2) IO will cause extorsion

   b. Adduction

      (1) SO will cause depression

      (2) IO will cause elevation



3. Clinical correlate: Extraocular muscle assessment

   a. Horizontal eye movements ↔

      (1) If the eye does not abduct: LR → lateral rectus

      (2) If the eye does not adduct: MR → medial rectus

   b. Vertical eye movements

      (1) Adducted position ↕

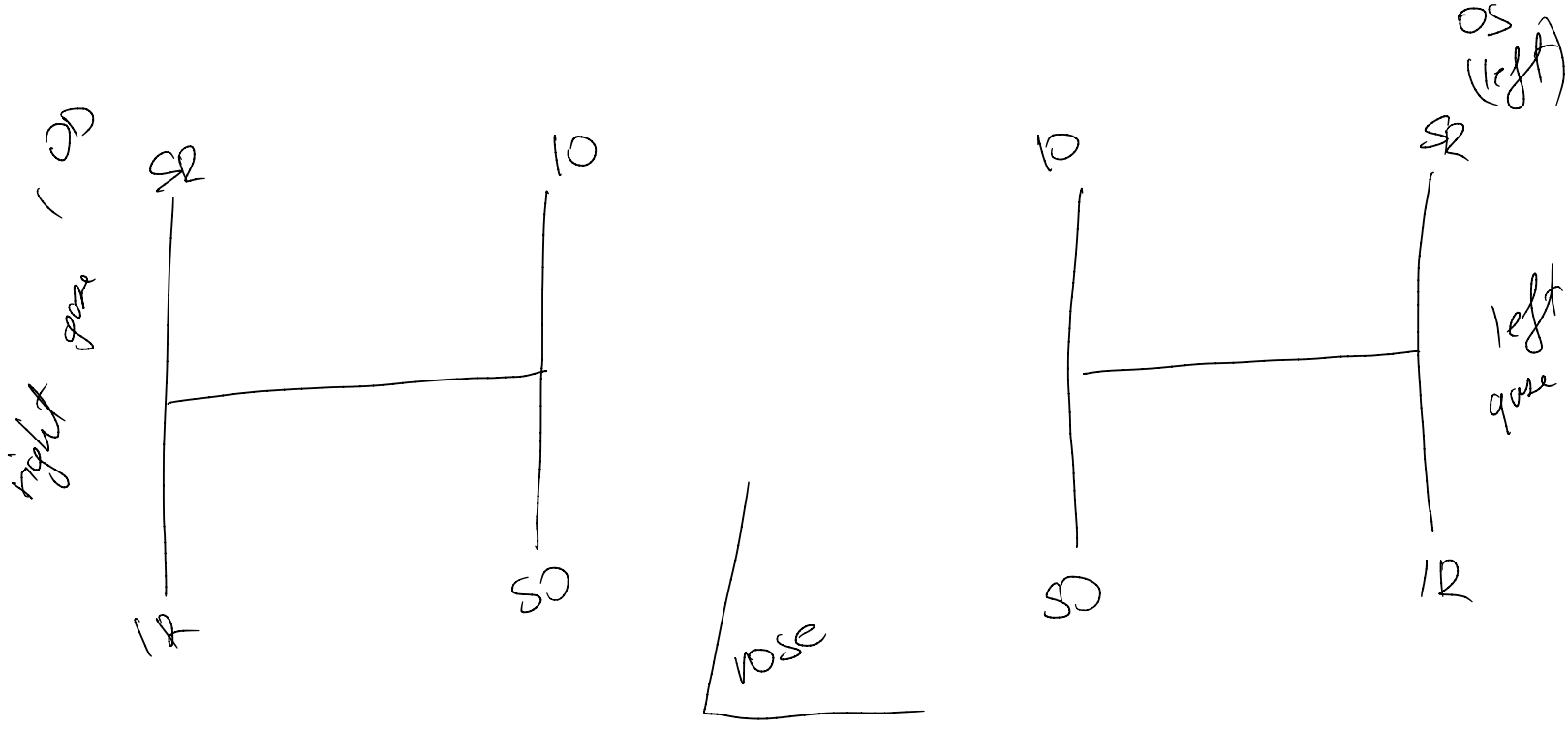
         i. If the eye does not elevate: IO – inferior oblique

         ii. If the eye does not depress: SO – superior oblique

      (2) Abducted position

         i. If the eye does not elevate: SR – superior rectus

         ii. If the eye does not depress: IR – inferior rectus



IV. Muscle coordination

A. Synergist/agonist muscles work together to accomplish the same action.

same

SR + IO → elevate ⇒ x-axis ⇒ agonist

↳ intorsion → extort ⇒ y-axis ⇒ antagonist

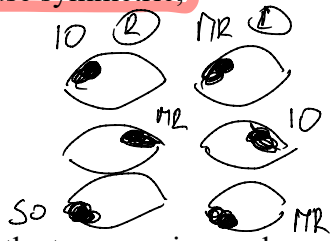
↳ adduct → abduct ⇒ z-axis ⇒ ″

B. Antagonist muscles work in opposite directions.

1. Sherrington's law of reciprocal innervation: contraction of one muscle is accompanied by a simultaneous and proportional relaxation of the antagonist muscle.

C. Yoke muscles: the muscles of the two eyes that work together to cause symmetric, binocular eye movements.

LR right eye + MR left eye



1. Hering's law of equal innervation: innervation to the muscles of the two eyes is equal and simultaneous.

